

$$(8) L = \{a^i b^j c^k \mid k = |i - j|\}$$

$$\begin{aligned} i > j &\rightarrow i = j + k \\ i < j &\rightarrow j = i + k \end{aligned}$$

$$S \rightarrow P \mid Q$$

$$i > j \left\{ \begin{aligned} P &= aPc \mid X \\ X &= aXb \mid \lambda \end{aligned} \right.$$

$$i < j \left\{ \begin{aligned} Q &= YZ \\ Y &= aYb \mid \lambda \\ Z &= bZc \mid \lambda \end{aligned} \right.$$

$$(9) L = \{a^i (b+c)^j d^k \mid i+k > j\}$$

$$S \rightarrow \lambda \mid APQ \mid PQD \mid APQD$$

$$\begin{aligned} P &\rightarrow aPZ \mid \lambda \\ Q &\rightarrow ZQd \mid \lambda \end{aligned} \quad \left. \vphantom{\begin{aligned} P &\rightarrow aPZ \mid \lambda \\ Q &\rightarrow ZQd \mid \lambda \end{aligned}} \right\} PQ \quad i+k > j$$

$$A \rightarrow aA \mid a$$

$$D \rightarrow dD \mid d$$

$$Z \rightarrow b \mid c$$

$$(10) L = \{a, b, c\}^* \mid N(c) = N(a) + N(b) \text{ and } N(a) \geq 1\}$$

$$S = AcAaA \mid AaAcA$$

$$A = AaAcA \mid AcAaA \mid \lambda \mid B$$

$$B = BbBcB \mid BcBbB \mid \lambda \mid A$$

Reduction:

$$S = XcXaX \mid XaXcX$$

$$X = XaXcX \mid XcXaX \mid XbXcX \mid XcXbX \mid \lambda$$